

Agentic AI — Introduction

Session Notes — AI Engineering Course

Topics: What is an Agent · Agent Loop · ReAct · Tools · Conceptual Terms · LangGraph Roadmap

❖ 1. What is an AI Agent?

An AI Agent is an **autonomous, intelligent system** that can:

- **Think** — reason about a goal using an LLM
- **Plan** — break down the task into steps (optional)
- **Act** — call tools to do things in the real world
- **Observe** — check the result of the action
- **Decide** — repeat the loop or stop when goal is achieved

LLM = the brain of the agent. All reasoning and decisions come from the LLM.

Tools = the hands of the agent. They let it act in the real world.

Real Example from Session

User goal: "Analyze Tesla and prepare an investment research report"

Step 1 → Think: I need current financial data

Step 2 → Act: Web search for Tesla financials

Step 3 → Observe: Got search results

Step 4 → Think: I need historical stock data

Step 5 → Act: Call financial API

Step 6 → Observe: Got stock data

Step 7 → Think: I need to calculate metrics

Step 8 → Act: Run calculations

Step 9 → Observe: All data collected → compile report

Step 10 → Done: Return final report to user

❖ 2. The Agent Loop

User Input / Goal

↓

Reasoning (LLM thinks about what to do)

↓

Planning (if needed – break task into steps)

↓

Action (call a tool / function)

↓

Observe (check the result)

↓

Decide (goal achieved? → stop : loop again)

Key: The loop runs until the agent gets the required final answer.

❖ 3. ReAct — Reason + Act

ReAct = Reasoning + Acting — a specific agent pattern.

Re = Reasoning — LLM thinks about what to do next

A = Action — call a tool or take a step

ct = Observation — observe the result of the action

Reason → Act → Observe → Reason → Act → Observe ...

until final answer found

Your mentor said: *ReAct is a subset of the bigger agent flow.*

The full agent loop can include planning, reflection, and more.

❖ 4. Tools — The Hands of the Agent

Tools are **functions** the LLM can call to take action in the real world.

Without tools, the LLM can only think and generate text.

With tools, it can search the web, call APIs, run code, query databases.

Common tools:

- **Web search** — find current information
- **Calculator** — perform mathematical operations
- **API calls** — get stock data, weather, news
- **Code executor** — run Python code
- **File reader** — read documents or databases
- **Email/Calendar** — interact with apps

❖ 5. Conceptual Terms to Know

Planning

Breaking a big goal into smaller steps. Can be static (fixed) or dynamic (adapts as it goes).

Task Decomposition

Splitting a complex task into sub-tasks that can be solved independently.

Reflection

Agent evaluates its own output — 'was this answer good enough?'

Self-Correction

Agent identifies its own mistakes and fixes them without human input.

Dynamic Planning

Plan changes based on intermediate results — not fixed upfront.

Execution

Actually running the planned steps — calling tools, generating output.

❖ 6. What's Coming — LangGraph Topics

Your mentor's plan for the next month (from 22nd August):

- **LangGraph architecture** — routing, parallelism, fan-in, fan-out
- **Dynamic vs deterministic workflows**
- **Persistence & checkpointing** — save agent state
- **Memory** — short-term and long-term agent memory
- **Human-in-the-loop** — when agent asks human for input
- **Streaming** — stream agent output in real time
- **Multi-agent systems**
- **MCP** — Model Context Protocol

→ **Evaluation & guardrails**

→ **Agentic RAG** — combining RAG with agent architecture

After agents: **end-to-end projects** + deployment

❖ 7. Quick Reference

AI Agent = autonomous system that thinks, plans, acts, observes, loops

LLM = brain of the agent — all reasoning comes from here

Tools = functions the agent calls to act in the real world

ReAct = Reason → Act → Observe (loop until answer found)

Planning = breaking goal into steps (optional, can be dynamic)

Reflection = agent checking its own output quality

Self-correction = agent fixing its own mistakes

LangGraph = framework for building agent workflows (next sessions)

Agentic RAG = RAG system with agent-like decision making

★ **Rule: Agent = LLM (brain) + Tools (hands) + Loop (behaviour)**

❖ 8. Your Mini-Project

Build the simplest possible agent — Plan then Act:

```
from openai import OpenAI

client = OpenAI(api_key='your-key')
```

```

def simple_agent(task):

    # Step 1 – Reason: break into steps

    plan = client.chat.completions.create(

        model='gpt-4o-mini',

        messages=[

            {'role':'system','content':'Break task into 3 steps.'},

            {'role':'user','content': task}

        ]).choices[0].message.content

    print(f'Plan:\n{plan}\n')


    # Step 2 – Act: execute step 1

    result = client.chat.completions.create(

        model='gpt-4o-mini',

        messages=[

            {'role':'system','content':'Execute step 1 only.'},

            {'role':'user','content':f'Task:{task} Plan:{plan}'}

        ]).choices[0].message.content

    return result


print(simple_agent('What is an AI agent?'))

```

This is the ReAct loop in its simplest form — Reason (plan) → Act (execute).

Once it works, try adding a third step: Observe and decide if the answer is complete.